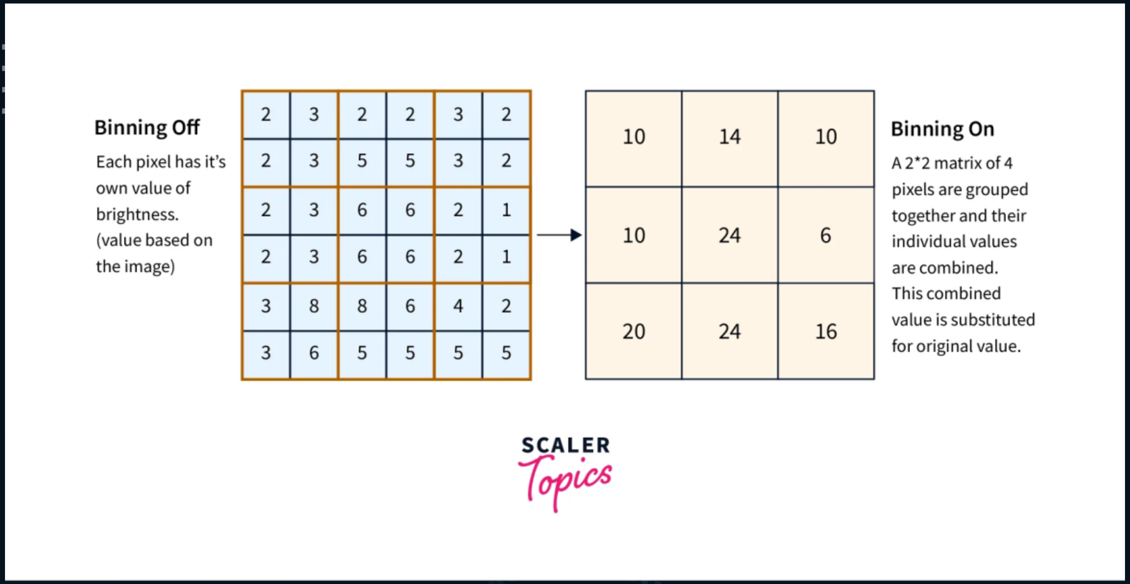
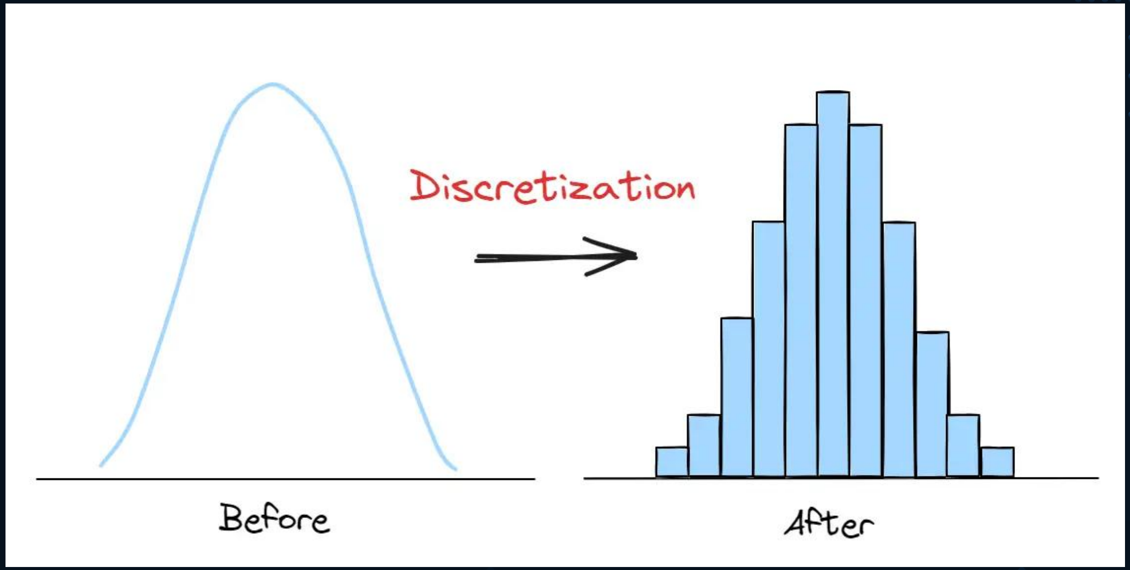
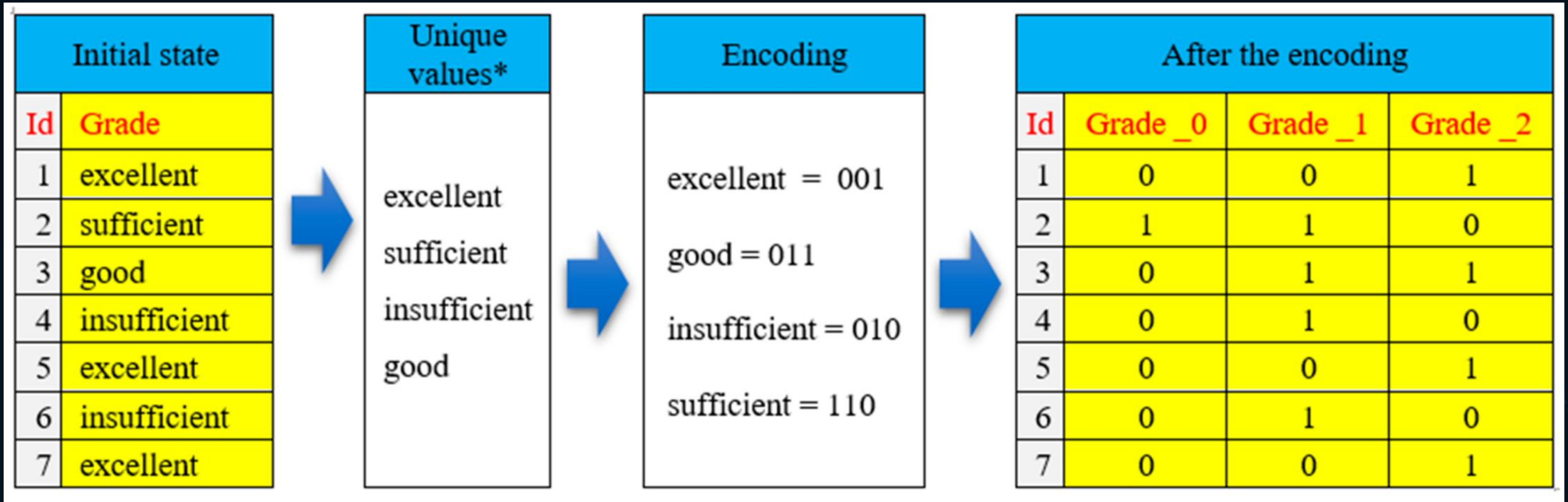




Binning & Discretization (From Scalars to Groups)

- **The Concept**: Reducing the "noise" of continuous numbers by grouping them into meaningful bins.
- **Why it works**: Sometimes the exact value (e.g., Age 22 vs 23) is less important than the category (e.g., Student vs Professional).
- **Types**:
  - **Equal Width**: Bins of equal size.
  - **Equal Frequency (Quantiles)**: Bins with the same number of data points.



Initial state	
Id	Grade
1	excellent
2	sufficient
3	good
4	insufficient
5	excellent
6	insufficient
7	excellent

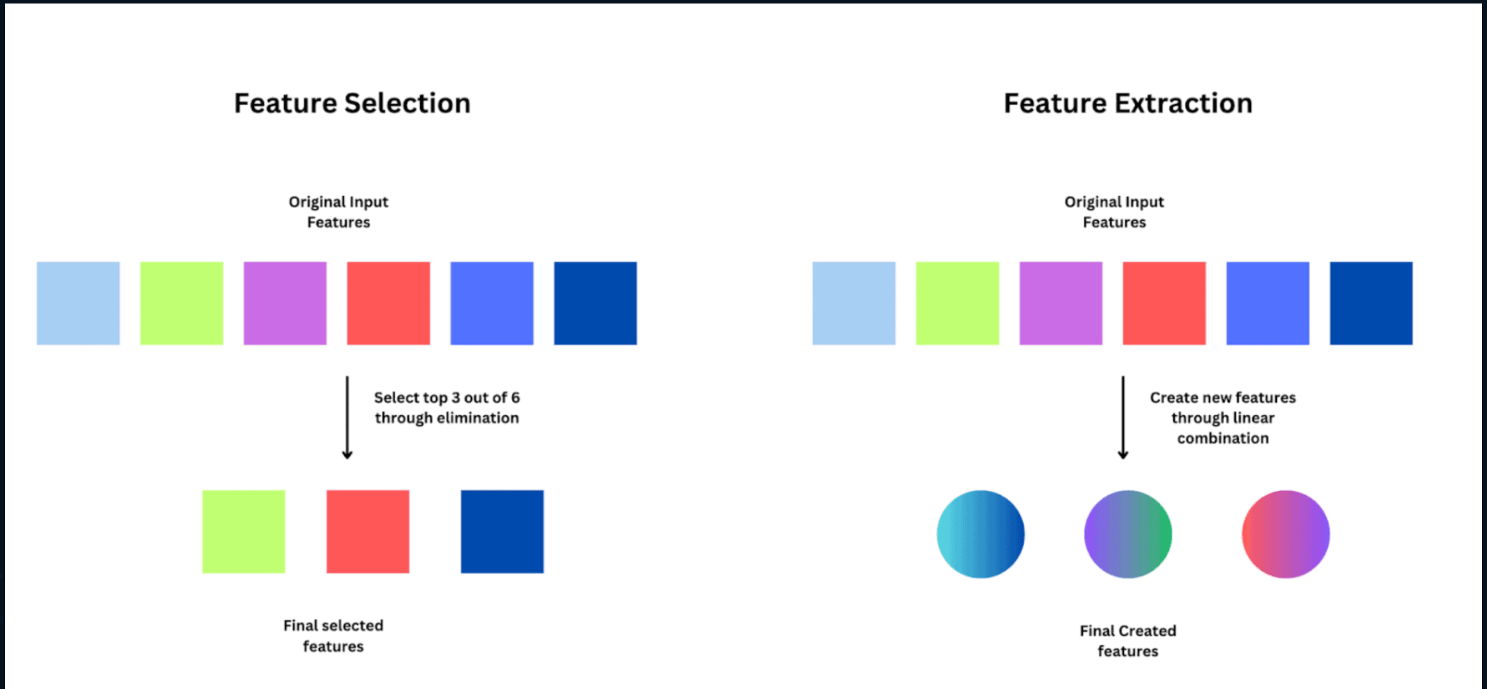
Unique values*
excellent
sufficient
insufficient
good

Encoding
excellent = 001
good = 011
insufficient = 010
sufficient = 110

After the encoding			
Id	Grade _0	Grade _1	Grade _2
1	0	0	1
2	1	1	0
3	0	1	1
4	0	1	0
5	0	0	1
6	0	1	0
7	0	0	1

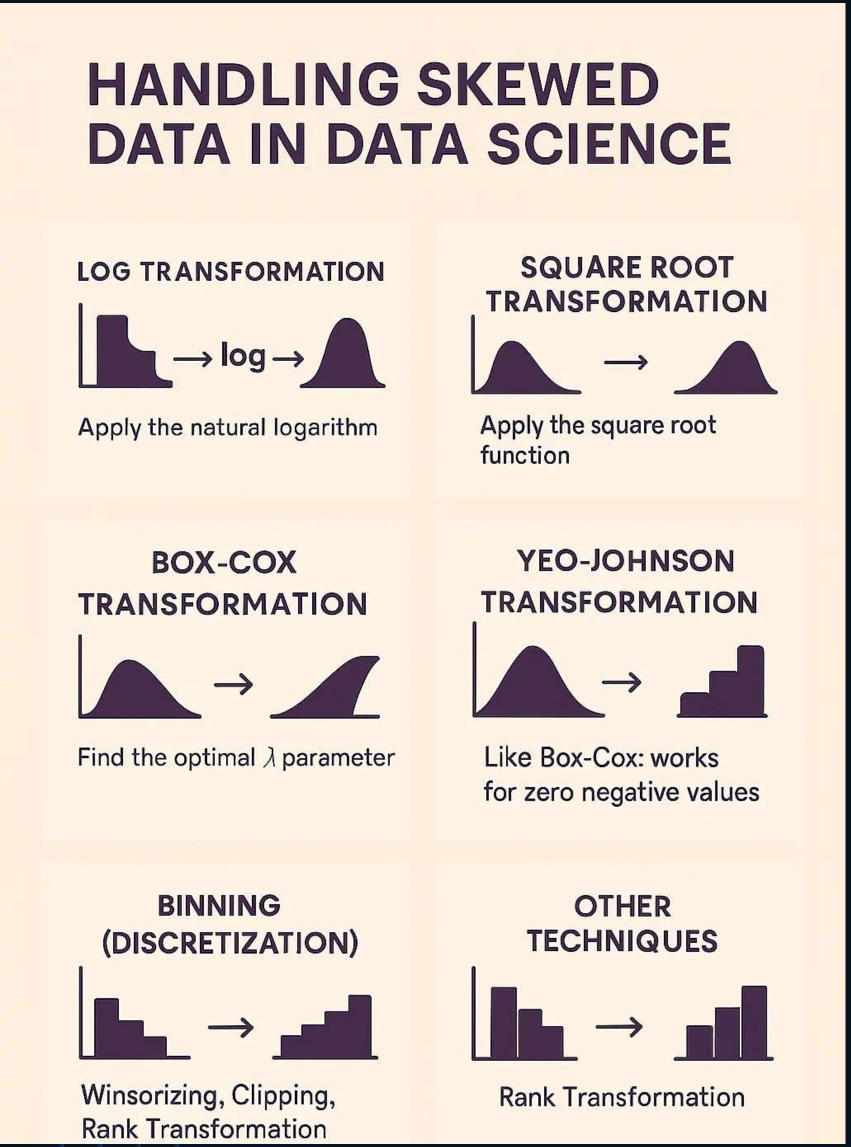
Encoding Categorical Data (Talking to the Machine)

- **The Problem**: Algorithms only understand math, not words like "Peshawar" or "Medical."
- **Label Encoding**: Assigning a unique integer to each category (0, 1, 2...). Best for "Ordinal" data where order matters (Small, Medium, Large).
- **One-Hot Encoding**: Creating a new binary column (0 or 1) for every unique category. Best for "Nominal" data where no order exists (City names).



Feature Extraction (Synthesizing New Insights)

- **Definition**: Creating new variables from existing ones to expose hidden patterns.
- **Date/Time Extraction**: Breaking a "Timestamp" into 'Hour of Day', 'Day of Week', or 'Is Weekend'.
- **Mathematical Extraction**: Creating a 'Price per Square Foot' column from 'Total Price' and 'Area'.
- **Domain Specific**: In medical imaging, extracting the 'Aspect Ratio' of a detected cell.



Handling Skewed Data (The Log Transform)

- **The "Bunching" Problem**: Real-world data (like wealth or rainfall) is often skewed, with most values bunched on the left and a few extreme values on the right.
- **Log Transformation**: Applying log(x) "pulls in" the extreme outliers and spreads out the bunched values.
- **The Result**: A distribution that looks "Normal" (Bell Curve), which is a requirement for many statistical models.